

Species Distribution Models

Foundations and applied modelling in R



What does a species distribution model do?



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It links observed records to environmental conditions, then projects that relationship across space.

Records

locations where the
species was observed



Environment

temperature, rainfall,
seasonality, habitat



Suitability

a mapped
hypothesis,
not proof of presence

Suitability is not the same as distribution



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Species may fail to occupy all suitable habitat, and records may miss places where the species occurs.

- Abiotic conditions can limit survival or reproduction.
- Biotic interactions can shape realised distributions.
- Dispersal barriers can prevent occupation of suitable environments.
- Sampling effort changes what we observe.

SDM output

environmental suitability

a hypothesis for
interpretation

Worked example: *Aedes aegypti* in Brazil

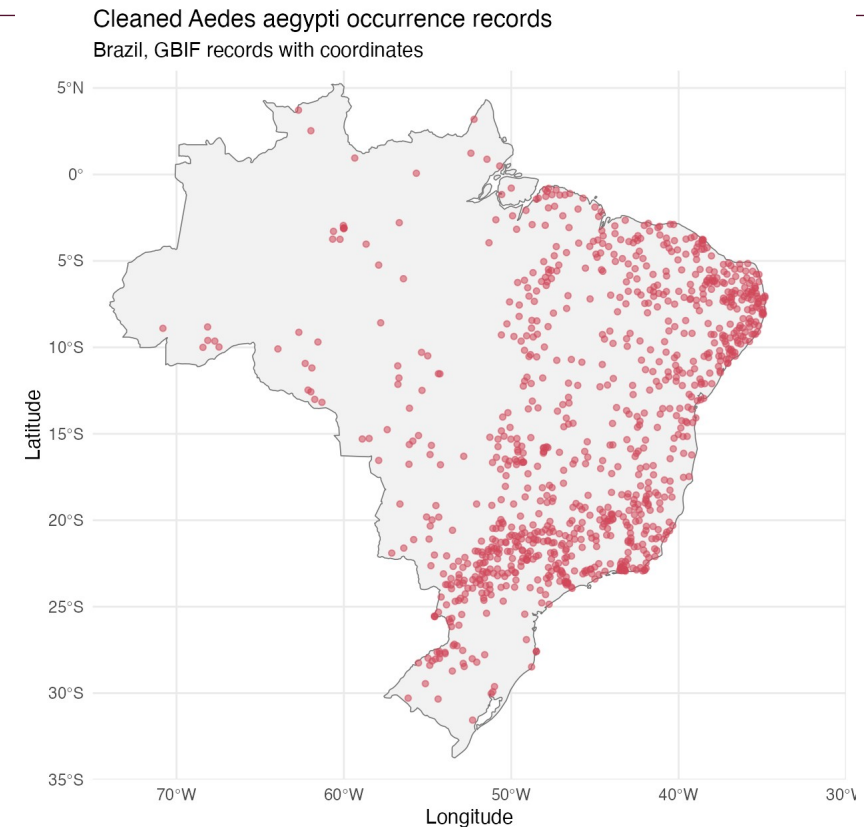


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After the foundation, the workshop uses a dengue vector as a practical modelling case.

- GBIF gives reusable occurrence records.
- Brazil provides a national-scale projection exercise.
- The model estimates vector suitability, not dengue risk.
- The same script can be adapted to other species.



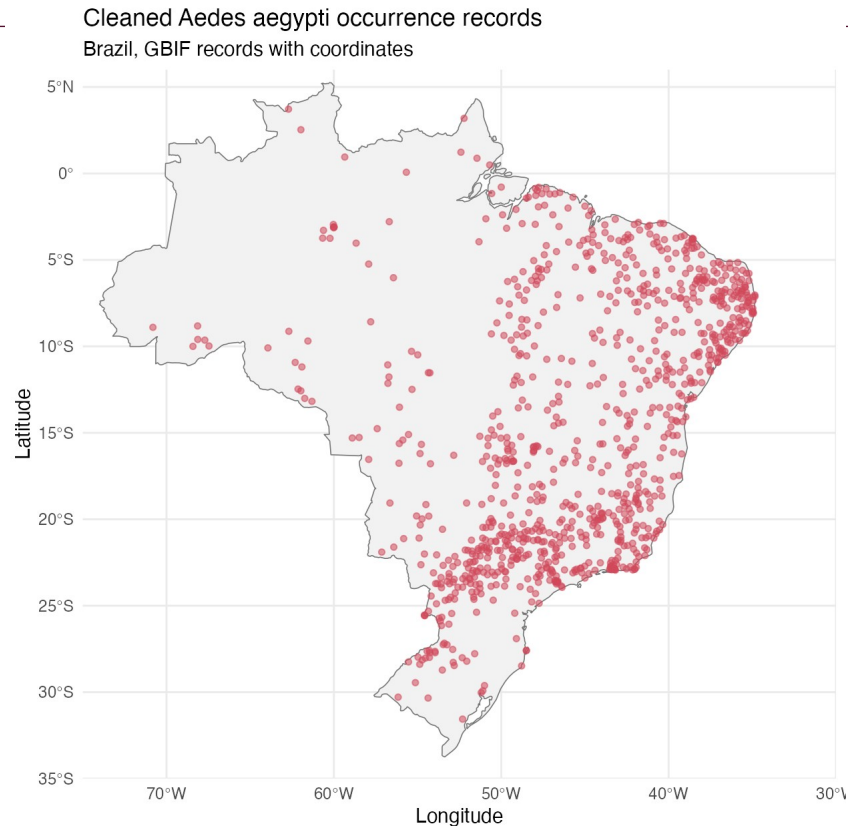
Occurrence data: useful, messy, teachable



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The first modelling decision is whether the records are good enough for the question.



Before modelling

- Remove records without coordinates.
- Check the study boundary.
- Thin duplicate raster cells.
- Keep a clean backup for live teaching.

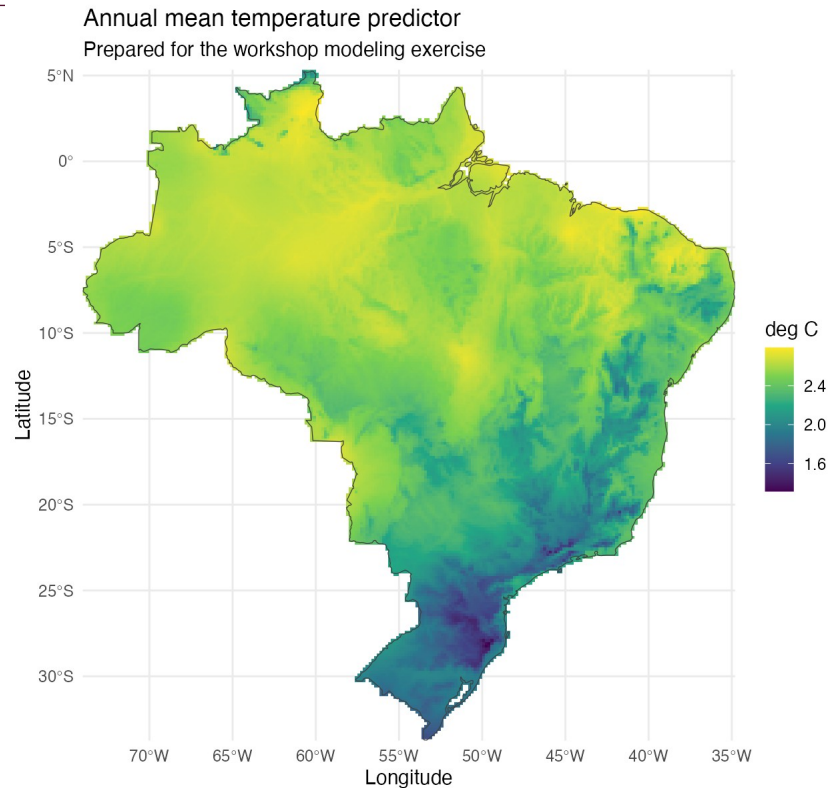
Predictors define the environmental lens



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We keep four climate predictors so the first model is fast enough and interpretable enough for a 90-minute workshop.



Workshop predictors

- Annual mean temperature
- Temperature seasonality
- Annual precipitation
- Precipitation seasonality

Presence-only data require pseudo-absences



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Pseudo-absences are comparison points. They are not confirmed absences.

Presences

known occurrence
records



Pseudo-absences

background contrast



Formatted data

ready for biomod2

In the script: `BIOMOD_FormatingData()`

Three algorithms, three modelling assumptions



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We use multiple algorithms to make model behaviour visible.

GLM

**transparent
statistical trend**

simpler shape

GBM

**boosted sequence
of trees**

flexible shape

RF

**many trees voting
together**

robust shape

Evaluation asks whether predictions generalise



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Good scores make the model worth inspecting. They do not make the map true.

Training data



Model



Test data

TSS

threshold-based balance of
sensitivity and specificity

AUCroc

threshold-independent ranking
of presences above absences

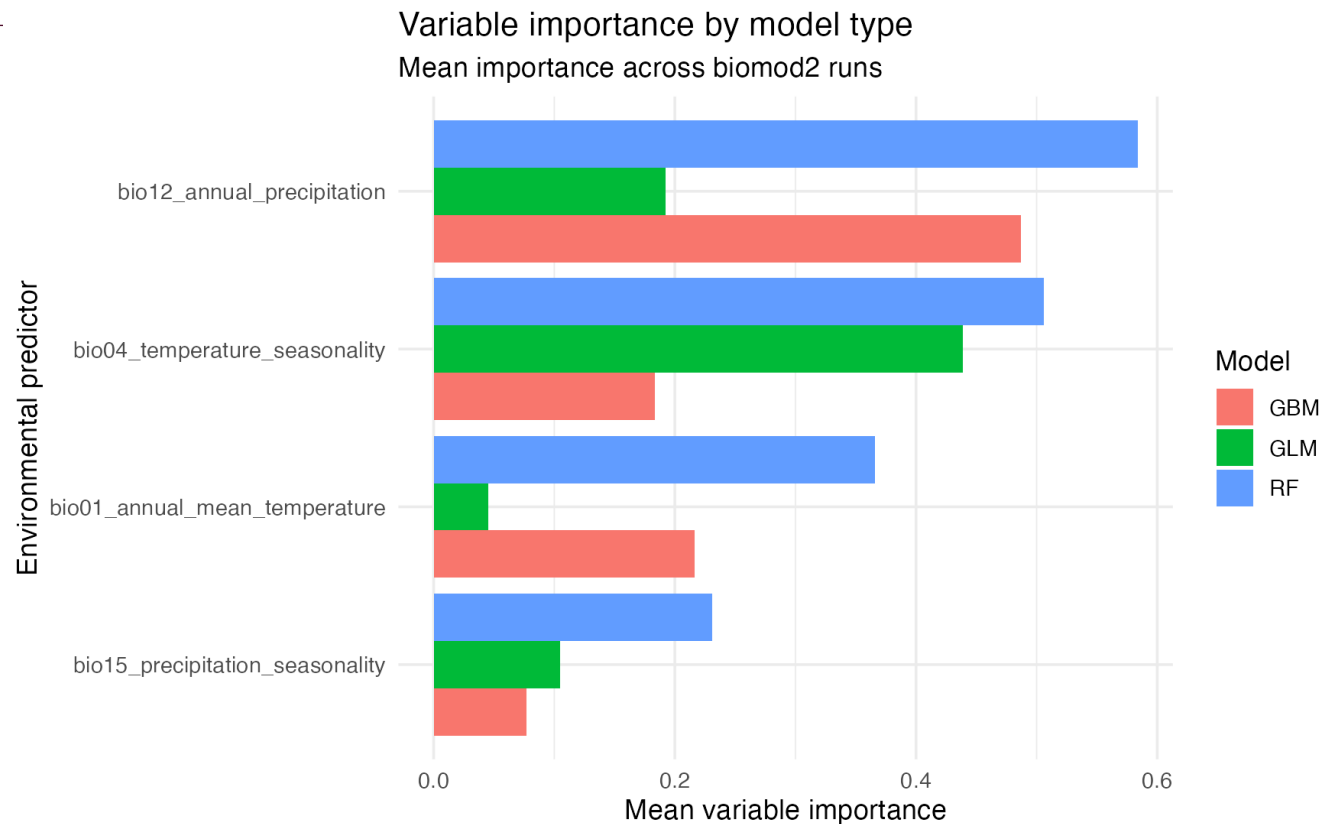
Variable importance: from output to questions



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Importance is model sensitivity, not biological proof.



Ask

- Which predictors changed predictions most?
- Do GLM, GBM, and RF agree?
- Could this be ecology, sampling, or correlation?

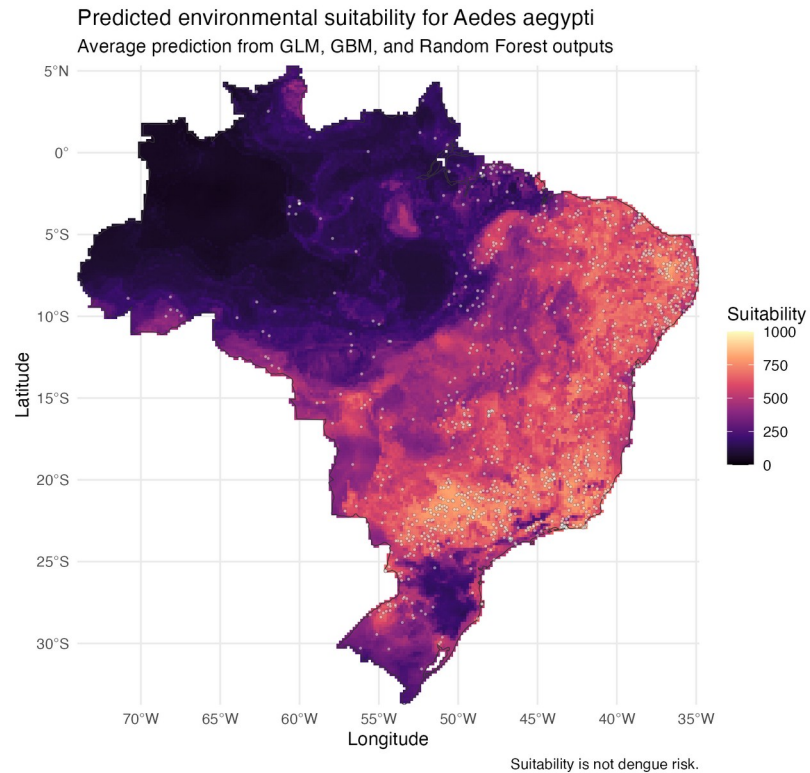
Ensembles and projections make choices visible



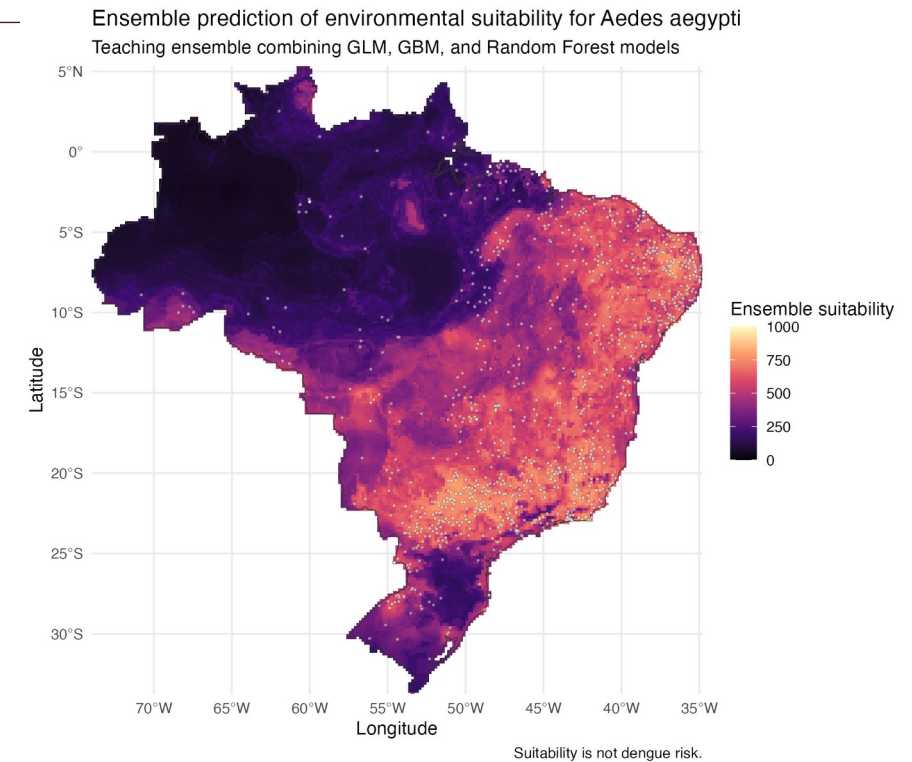
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The map is a synthesis of model behaviour across Brazil.



individual model average



ensemble projection

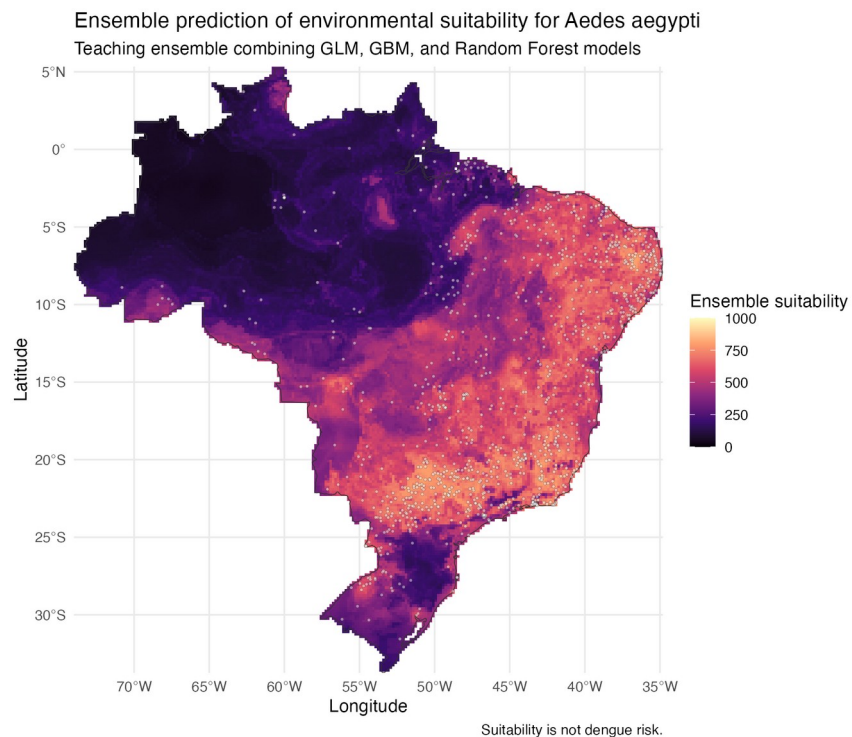
A good SDM should sharpen the next conversation



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Now we build the model line by line in R.



Before acting on a map, ask:

- Where is suitability high or low?
- Where are the data thin?
- What could be sampling bias?
- Which predictor story is plausible?
- What dengue surveillance data would you add?